BOARD OF SECONDARY EDUCATION, KARACHI

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2024

PHYSICS

Past Paper 2024

Retyped faithfully from the original board paper — same questions, no changes.

SECTION "A" MULTIPLE CHOICE QUESTIONS (MCQs)

Maximum Marks: 12

Note: (i) Attempt all the Questions of this Section. (ii) Do not copy down the part questions. Write only the answer against the proper number of the Question and its part according to the question Paper. (iii) Each question carries 1 Marks

Q1. 1. Choose the correct Answer for each from the given question:

Q1. Spring balance is used to measure:

(A) Mass
(B) Weight
(C) Elasticity
(D) Density

Q2. A burning candle is an example of this state of matter:

(A) Gas
(B) Liquid
(C) Solid
(D) All of these

Q3. Evaporation occurs at:

(A) Freezing point
(B) Melting point
(C) Boiling point
(D) All temperatures

Q4. It is not a unit of pressure:

(A) Pascal
(B) Bar
(C) Atmosphere
(D) Newton

Q5. Centrifugal force is always:

(A) towards the centre
(B) away from centre
(C) towards the circular path
(D) all sides

Q6. The least distance covered in a particular direction is called:

(A) Distance
(B) Speed
(C) Displacement
(D) Acceleration

Q7. 91.60 has significant numbers:

(A) 2
(B) 3
(C) 4
(D) 5

Q8. It is not a vector quantity:

(A) Time
(B) Weight
(C) Displacement
(D) Acceleration

Q9. Natural satellite of Earth is:

(A) Neptune
(B) Jupiter
(C) Moon
(D) Mars

Q10. If a machine performs 20J work in 10 seconds. Then its power will be:

(A) 200 watt
(B) 20 watt
(C) 2 watt
(D) 0.2 watt

Q11. 1.013 × 105 Pa is equal to:

(A) 1 atmosphere
(B) 2 atmosphere
(C) 3 atmosphere
(D) 4 atmosphere

Q12. A body of mass 50 kilogram whose acceleration is 5ms−2, the force affecting on it will be:

(A) 200 N
(B) 250 N
(C) 260 N
(D) 270 N

SECTION “B” (SHORT-ANSWER QUESTIONS)

Maximum Marks: 24

> Note: Note: Answer any EIGHT questions from this section. Each question carries Three marks.

Q2. Define fundamental and derived quantites. Also write three examples of each.

Q3. State Newton's Second Law of Motion and derive its equation.

Q4. Write down three difference between Boiling Point and Evaporation.

Q5. What is potential energy? Derive P.E = mgh.

Q6. What do you mean by elasticity? State Hooke's Law and write its mathematical expression.

Q7. Define torque. Write down its formula and unit.

Q8. A bus starts form rest and travels along a straight path its velocity becomes 15ms−1 in 5 seconds. Calculate acceleration of the bus.

Q9. A car is running on a circular path of highway having about 100m radius. The mass of the car is 600kg and its velocity is 72kmh−1. Find centripetal force exerted by the car.

Q10. Calculate the spring constant which extends by a distance of 3.5cm when a load of 14N hung from its end.

Q11. A planet has mass four times of earth and radius two time that of earth is 10 ms−2. What will be its acceleration due to gravity on the planet?

Q12. What will be the kinetic energy of a boy of mass 50kg driving a bike with the velocity of 2ms−1

Q13. A man is pushing a Trolley for a horizontal ground with a force of 100N making an angle of 60° with ground. Find the horizontal and vertical components of the force.

SECTION “C” (DETAILED-ANSWER QUESTIONS)

Maximum Marks: 24

> Note: Note: Attempt any FOUR questions from this Section. Each question carries (Six) 6 marks.

Q14. What is kinematics? Derive the equation $2aS = vf2 - vi2$

Q15. Define equilibrium. Explain its types and two conditions.

Q16. State and explain Boyle's Law. Write its one application.

Q17. Calculate the mass of earth by using Newton's law of gravitation.

Q18. Define Thermal expansion. Derive $\Delta L = \alpha L \Delta T$

Q19. Write any two advantages and two disadvantages of friction. Also write two methods of reducing friction.